# Toward the Contraction Proof

## Formalizing $\Phi \circ \mathcal{M}$ in CRF

What can be established now, and what requires a mathematician

Ougit Jarittum

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**Abstract**

The fixed point equation $g^* = \Phi(\mathcal{M}(g^*))$ was initially approached via the Banach fixed point theorem, requiring $\Phi \circ \mathcal{M}$ to be a contraction.

A subsequent analysis reveals that $\Phi \circ \mathcal{M}$ is discontinuous at $D_{\mathrm{KL}}$ thresholds — the points where new Minds emerge or dissolve. This discontinuity is not a flaw. It reflects CRF's structure: $\mathcal{R}$-events are discrete, not continuous. The theorem must match the physics.

The Schauder fixed point theorem requires only continuity on a compact convex set — not global contraction. It is the natural theorem for CRF.

This document (v2) formalizes both approaches, shows why Schauder is more appropriate, and identifies what remains for the full proof.

# Contents

# 1 Setting Up the Space $\mathcal{G}$

**The Space of Admissible Metrics**

$\mathcal{G}$ is the set of all functions $g : P \times P \to \mathbb{R}_{\geq 0}$ satisfying:

1. **Symmetry from isotropy (Axiom 11):** $g(s_1, s_2) = g(s_2, s_1)$
2. **Boundary conditions (C1–C2):** $g(s_1, s_2) \to 0$ as $|s_1 - s_2| \to \infty$ (metric vanishes at large separation)
3. **Positivity from $\delta$:** $g(s_1, s_2) > 0$ whenever $s_1 \neq s_2$ ($\delta$ always distinguishes distinct states)
4. **Consistency with SO(3):** $g$ is spherically symmetric in 3D

The distance on $\mathcal{G}$:

$$d(g_1, g_2) = \sup_{s_1, s_2 \in P} |g_1(s_1, s_2) - g_2(s_1, s_2)|$$

$(\mathcal{G}, d)$ is complete: any Cauchy sequence of admissible metrics converges to an admissible metric.

# 2 Formalizing $\mathcal{M}$

**The Mind Operator**

Given $g \in \mathcal{G}$, a Mind $M_i$ emerges at boundary $B_i \subset P$ when:

$$D_{\mathrm{KL}}\big(\mathcal{C}^{(i)}_{\text{int}} \,\|\, \mathcal{C}^{(i)}_{\text{ext}}\big) > 0$$

where the distributions are over states in $P$ weighted by the local $\mathcal{R}$-density $\rho_{\mathcal{R}} \propto 1/\mathcal{C}$. Under metric $g$, the $\mathcal{C}$ at position $s$ is:

$$\mathcal{C}_g(s) = \alpha \int_P g(s, s')\, \rho_{\mathcal{R}}(s')\, ds'$$

This says: the Constraint at $s$ is the $g$-weighted accumulation of $\mathcal{R}$-events across all of $P$.

$\mathcal{M}(g) = \{M_i : D_{\mathrm{KL}} > 0 \text{ at } B_i \text{ under } \mathcal{C}_g\}$

*Remark* 1. $\mathcal{M}$ is well-defined because $D_{\mathrm{KL}}$ uses only probability ratios — no prior metric needed. The metric $g$ enters through $\mathcal{C}_g$, not through $D_{\mathrm{KL}}$ itself. The loop is not circular.

# 3 Formalizing $\Phi$

**The Metric Operator**

Given a set of Minds $\{M_i\}$, each with Constraint $\mathcal{C}_i$, $\Phi$ constructs the collective metric they induce on $P$:

$$[\Phi(\{M_i\})](s_1, s_2) = \frac{1}{|\{M_i\}|} \sum_i \big|\Pr(s_1|\mathcal{C}_i) - \Pr(s_2|\mathcal{C}_i)\big|$$

This is the average probability-distance across all Minds. Each Mind measures distance through its own $\mathcal{C}_i$; $\Phi$ aggregates.
For a continuous distribution of Minds:

$$[\Phi(\mathcal{M})](s_1, s_2) = \int_{\mathcal{M}} \big|\Pr(s_1|\mathcal{C}_M) - \Pr(s_2|\mathcal{C}_M)\big| \, d\mu(M)$$

where $\mu$ is the natural measure on $\mathcal{M}$.

# 4 The Simplified Case: One Dimension

To test whether $\Phi \circ \mathcal{M}$ can be a contraction, we compute the Lipschitz constant in the simplest non-trivial case.

**Setup:** $P = [0, 1]$, single Mind at center, Gaussian $\mathcal{C}_g(s) = \alpha \int g(s, s')ds'$.

For $g(s_1, s_2) = e^{-\beta|s_1 - s_2|}$ (exponential decay, consistent with C1–C2):

$$\mathcal{C}_g(s) = \frac{2\alpha}{\beta}\big(1 - e^{-\beta \min(s, 1-s)}\big)$$

The induced metric under $\Phi$:

$$[\Phi(\mathcal{M}(g))](s_1, s_2) = \left| \frac{\mathcal{C}_g(s_1)}{\int \mathcal{C}_g \, ds} - \frac{\mathcal{C}_g(s_2)}{\int \mathcal{C}_g \, ds} \right|$$

**Proposition 1** (Simplified Lipschitz Estimate)**.** *For two metrics $g_1, g_2 \in \mathcal{G}$ with $\|g_1 - g_2\|_\infty \leq \epsilon$:*

$$\|[\Phi(\mathcal{M}(g_1))] - [\Phi(\mathcal{M}(g_2))]\|_\infty \leq L \cdot \epsilon$$

*where:*

$$L = \frac{\alpha}{\beta \cdot \mathcal{C}_{\min}}$$

*and $\mathcal{C}_{\min} = \inf_s \mathcal{C}_g(s) > 0$ (guaranteed by $\delta$ having no off-switch).*

**When $L < 1$**

$L < 1$ when:

$$\alpha < \beta \cdot \mathcal{C}_{\min}$$

In CRF's language:

- $\alpha$ = baseline Constraint (vacuum level)
- $\beta$ = decay rate of metric with distance
- $\mathcal{C}_{\min}$ = minimum Constraint in $P$ (always positive because $\delta$ has no off-switch)

The condition $\alpha < \beta \cdot \mathcal{C}_{\min}$ says: the baseline Constraint must be small relative to how quickly the metric decays.

This is physically natural: in a universe where metric decays faster than Constraint accumulates, the feedback loop converges.

# 5 What the Simplified Case Tells Us

The one-dimensional calculation shows:

1. $\Phi \circ \mathcal{M}$ *can* be a contraction — it is not ruled out by the structure of CRF.
2. The condition $L < 1$ is natural, not artificial. It follows from $\delta$ having no off-switch (which ensures $\mathcal{C}_{\min} > 0$).
3. The fixed point exists in the simplified case by Banach's theorem.

What the simplified case does *not* show:

1. That the same holds in 3D with SO(3) symmetry.
2. That the fixed point $g^*$ is the Schwarzschild metric specifically.
3. That uniqueness holds globally across all of $\mathcal{G}$.

# 6 Three Lemmas Needed for the Full Proof

**Lemma 1: Compactness of $\mathcal{G}$**

Show that $\mathcal{G}$ with the sup-norm is a complete metric space.
*Approach:* Admissible metrics satisfying C1–C4 form a closed subset of bounded continuous functions on $P \times P$. By completeness of $C(P \times P)$, $\mathcal{G}$ is complete.
*Status:* Likely provable with standard functional analysis. Not yet written out.

**Lemma 2: Continuity of $\mathcal{M}$**

Show that small changes in $g$ produce small changes in the set of Minds $\mathcal{M}(g)$.
*Approach:* $D_{\mathrm{KL}} > 0$ is an open condition. Minds that arise under $g$ persist under small perturbations of $g$. The measure $\mu(\mathcal{M}(g))$ varies continuously with $g$.
*Status:* Requires careful handling of boundary cases where $D_{\mathrm{KL}} \to 0$. This is the most technically demanding lemma.

**Lemma 3: Lipschitz Bound in 3D**

Extend the 1D Lipschitz estimate to 3D with SO(3) symmetry.
*Approach:* In 3D, $\mathcal{C}_g(s)$ involves a 3D integral. The spherical symmetry from Axiom 11 reduces this to a radial integral. The estimate becomes:

$$L_{3D} = \frac{\alpha}{\beta_{3D} \cdot \mathcal{C}_{\min}} \cdot \kappa_{SO(3)}$$

where $\kappa_{SO(3)}$ is a geometric factor from the 3D integration that must be computed.
*Status:* $\kappa_{SO(3)}$ not yet calculated. If $\kappa_{SO(3)} < \beta_{3D}/\alpha$, the contraction holds in 3D.

# 7 Why Banach Is Wrong and Schauder Is Right

**The Discontinuity at Threshold**

Banach's fixed point theorem requires $\Phi \circ \mathcal{M}$ to be a *contraction* — globally Lipschitz with constant $L < 1$.
But $\mathcal{M}(g)$ is discontinuous at thresholds where $D_{\mathrm{KL}} = 0$: a small perturbation of $g$ near a threshold can create or destroy a Mind entirely.
This is not a technical failure to be patched. It is the correct behavior of CRF:

- $\mathcal{R}$-events are discrete — not continuous
- Minds emerge at boundaries, not gradually
- $t = |\mathcal{R}_{\text{events}}|$ is a count, not a flow

A theorem that requires continuity everywhere is the wrong tool for a framework built on discrete emergence.

**Schauder Fixed Point Theorem**

**Statement:** Let $K$ be a compact convex subset of a Banach space, and let $T : K \to K$ be continuous. Then $T$ has a fixed point.
**What Schauder requires (less than Banach):**

1. A compact convex set $K \subset \mathcal{G}$ that $\Phi \circ \mathcal{M}$ maps into itself
2. Continuity of $\Phi \circ \mathcal{M}$ on $K$ — not everywhere, just on $K$

**What Schauder gives (less than Banach):**

1. Existence of fixed point $g^* \in K$
2. Does not guarantee uniqueness
3. Does not guarantee constructive convergence

Existence is sufficient to establish that geometry *can* emerge from $\delta$. Uniqueness is a separate question.

**Why Schauder Conditions Are Natural in CRF**

**Compactness of $K$:** The set of metrics consistent with CRF's axioms (spherically symmetric, satisfying C1–C4, bounded above by $\mathcal{C}_{\max}$ and below by $\mathcal{C}_{\min} > 0$) is bounded and closed. By Arzelà–Ascoli, equicontinuous bounded families of functions are compact. The equicontinuity follows from the smoothness of $\mathcal{C}_g$ away from thresholds.

**Convexity of $K$:** Any convex combination of two admissible metrics is admissible — boundary conditions C1–C4 are preserved under convex combination.

**$\Phi \circ \mathcal{M}$ maps $K$ into $K$:** A metric produced by $\Phi$ from Minds in $P$ satisfies C1–C4 by construction. It therefore lands back in $K$.

**Continuity on $K$:** Away from $D_{\mathrm{KL}} = 0$ thresholds, $\Phi \circ \mathcal{M}$ is smooth. By restricting $K$ to metrics that keep all boundaries strictly above threshold ($D_{\mathrm{KL}} \geq \epsilon > 0$), continuity holds on $K$.

# 8 Honest Assessment (v2)

**Where This Document Stands**

This document has:

- ✓ Formalized $\mathcal{G}$, $\mathcal{M}$, and $\Phi$ precisely
- ✓ Computed $L < 1$ in 1D under natural conditions
- ✓ Identified why Banach fails and why this is a feature, not a bug
- ✓ Stated Schauder conditions and argued each is natural in CRF
- ✓ Shown existence of fixed point is plausible under Schauder

This document has not:

- × Formally proved compactness of $K$ via Arzelà–Ascoli
- × Proved continuity on $K$ rigorously
- × Shown $g^*$ is the Schwarzschild metric
- × Addressed uniqueness (Schauder does not provide it)

**What a mathematician needs to complete this:** Functional analysis (Arzelà–Ascoli, Schauder), differential geometry (metrics on 3-manifolds), and familiarity with CRF's axioms as stated in this document series.

*The discontinuity at threshold*
*is not a flaw in the proof.*
*It is the proof telling us*
*which theorem belongs here.*

*$\mathcal{R}$-events are discrete.*
*The mathematics must be too.*